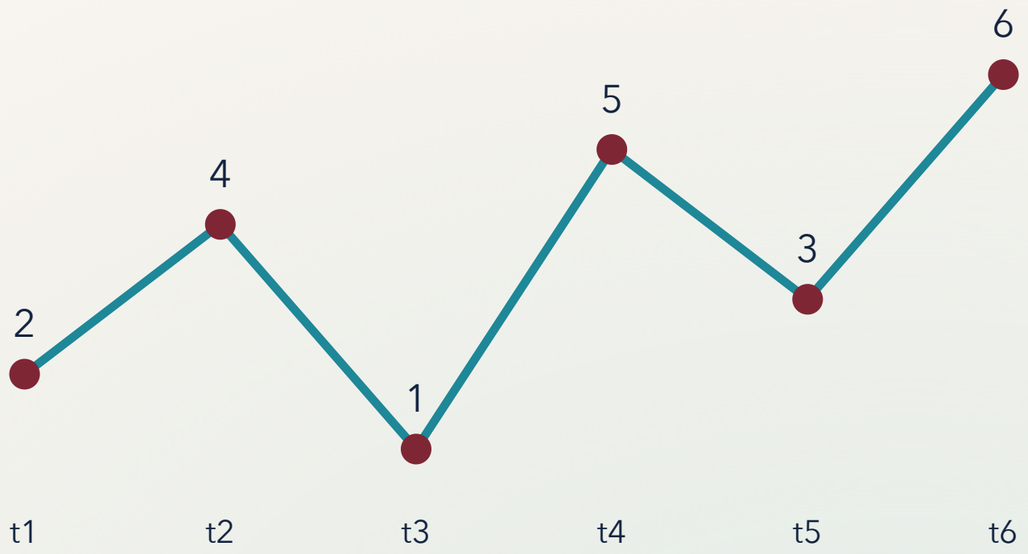
Order changes the learning problem

An observation becomes a sequence when its meaning depends on what came before it.

> $x_1, x_2, \ldots, x_t \rightarrow \text{predict } x_{t+1}$

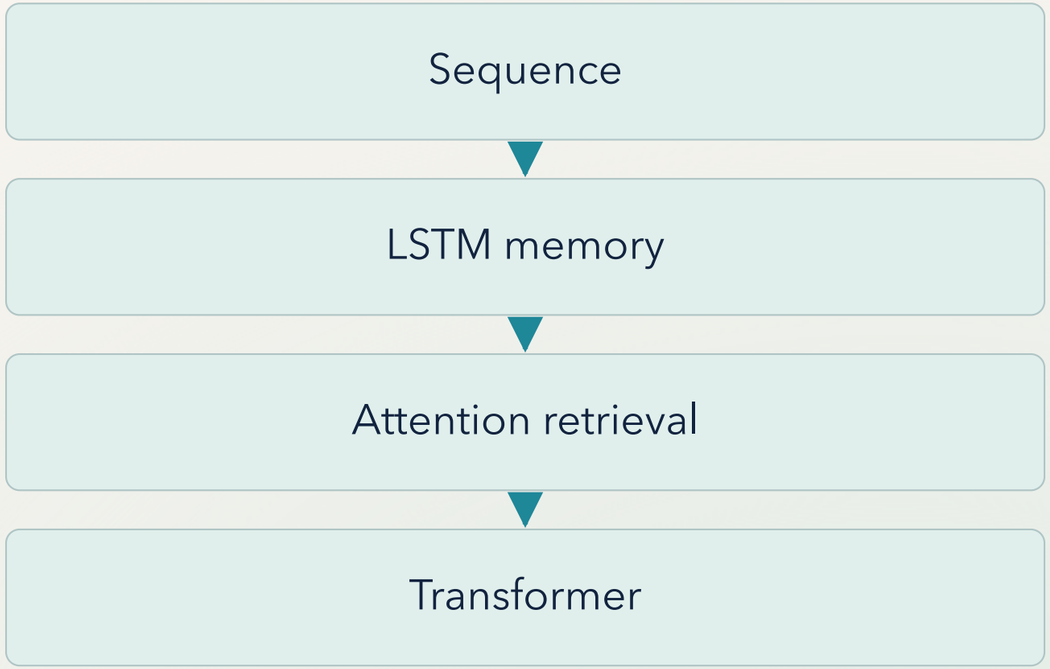


Order is part of the input

Mechanism diagram / illustrative teaching values

Three mechanisms, one learning path

LSTM stores a running state. Attention retrieves relevant states. A Transformer builds representations with attention.



Mechanism diagram / illustrative teaching values

The same values can tell different stories

A sequence contains both values and an ordering rule.

> $[1, 2, 4, 8] \neq [8, 1, 4, 2]$

What remains identical after shuffling?

A

$1 \rightarrow 2 \rightarrow 4 \rightarrow 8$

B

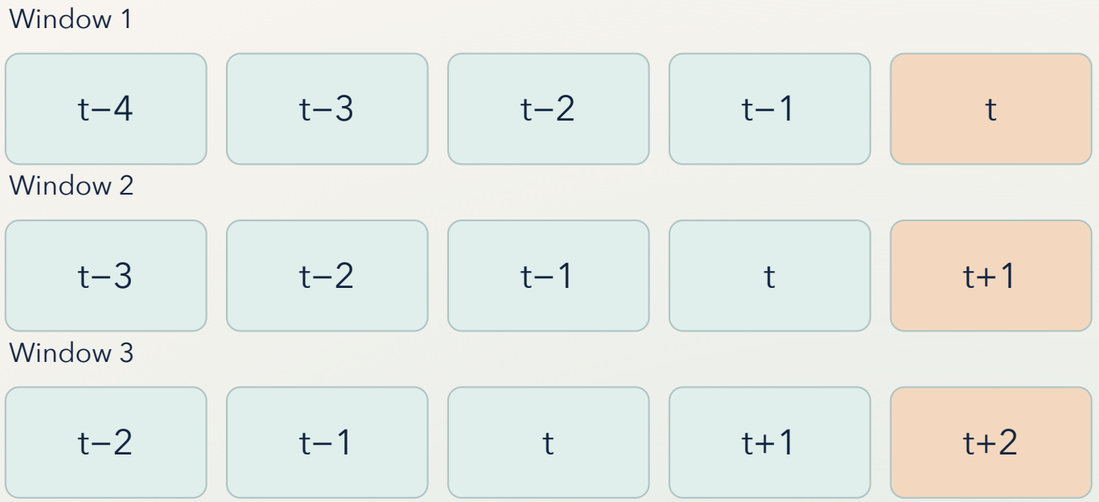
$8 \rightarrow 1 \rightarrow 4 \rightarrow 2$

Mechanism diagram / illustrative teaching values

Define the forecast origin

The prediction must use only information available at its origin.

> `history: x[t−L:t] target: x[t]`



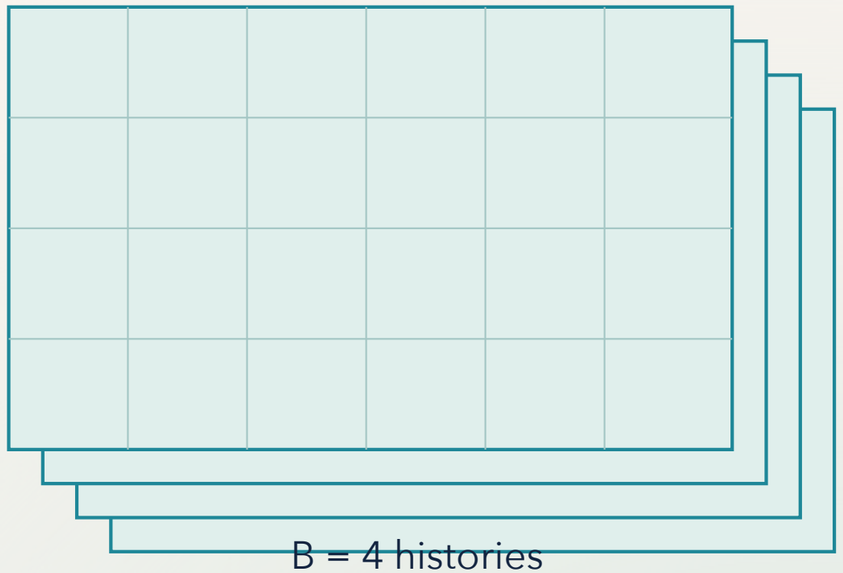
History shifts forward; target stays excluded

Mechanism diagram / illustrative teaching values

One sample is an entire window

Sequence models receive a batch of histories, with features at every time step.

$$X \in \mathbb{R}^{B \times T \times f}$$

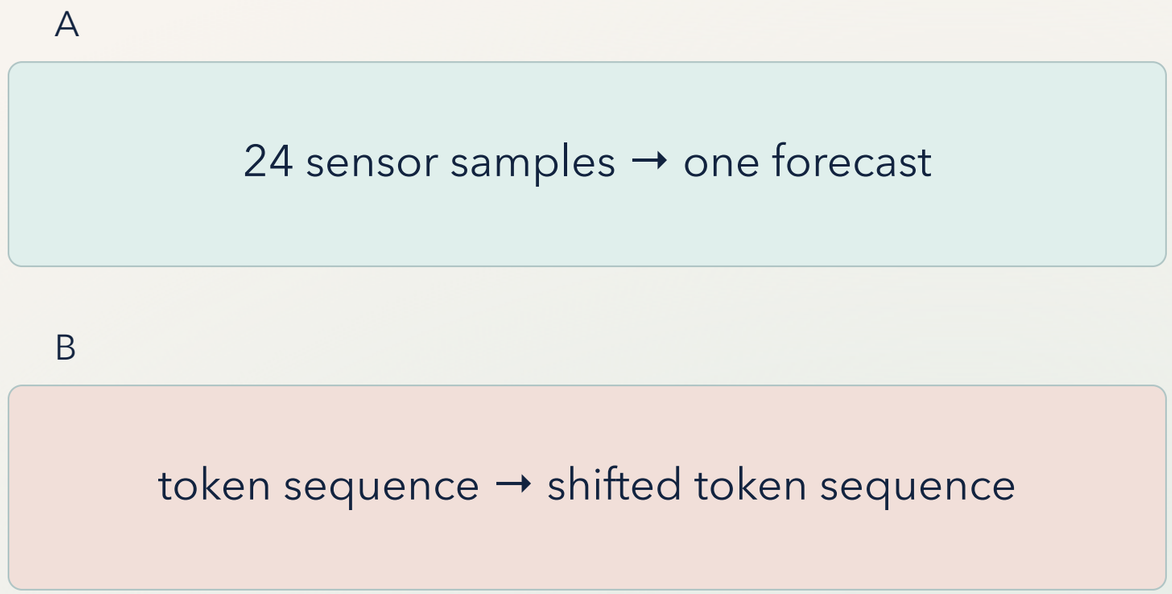


B = 4 histories
T = 24 samples
F = 1 sensor

Mechanism diagram / illustrative teaching values

Targets determine the output structure

Many-to-one predicts one value; many-to-many predicts a value at multiple positions.

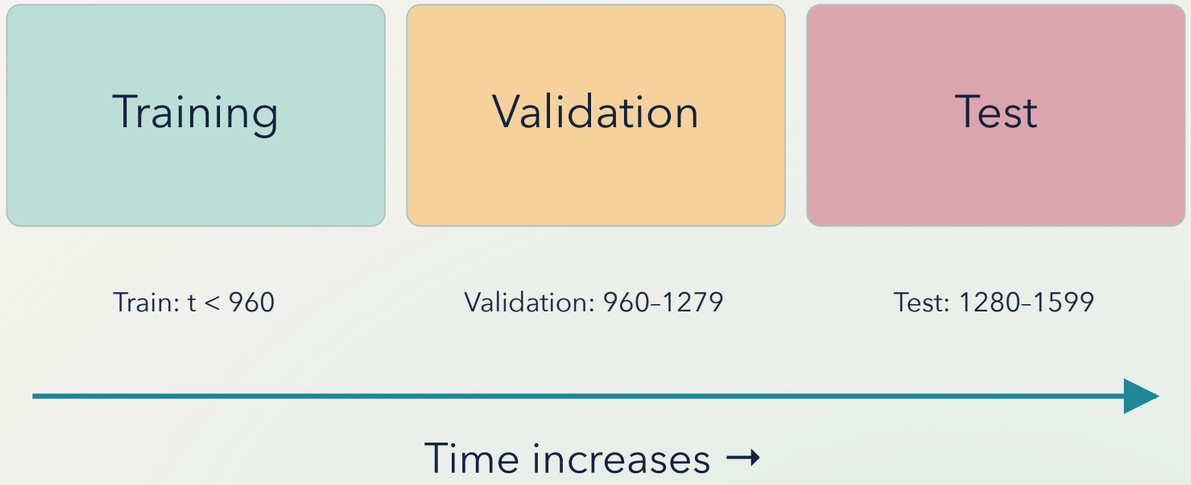


Mechanism diagram / illustrative teaching values

Split time before selecting a model

Train on earlier targets, select on later validation, then evaluate on test.

> `train < validation < test`

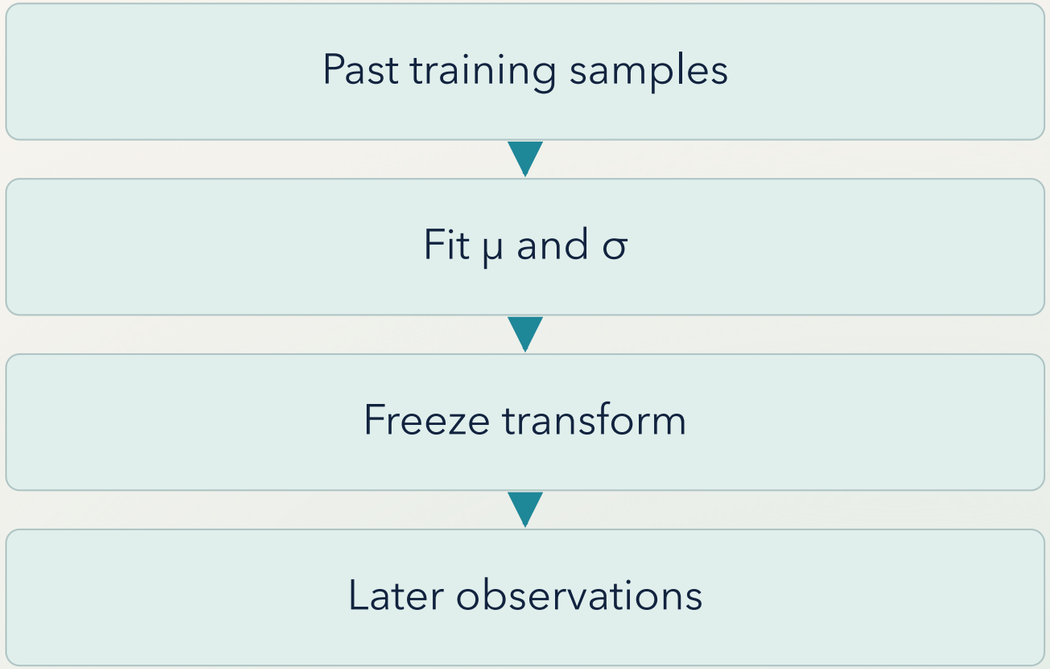


Mechanism diagram / illustrative teaching values

Fit normalization on the training past

Global means and variances can reveal future distribution information.

```
z = (x − μtrain) / σtrain
```

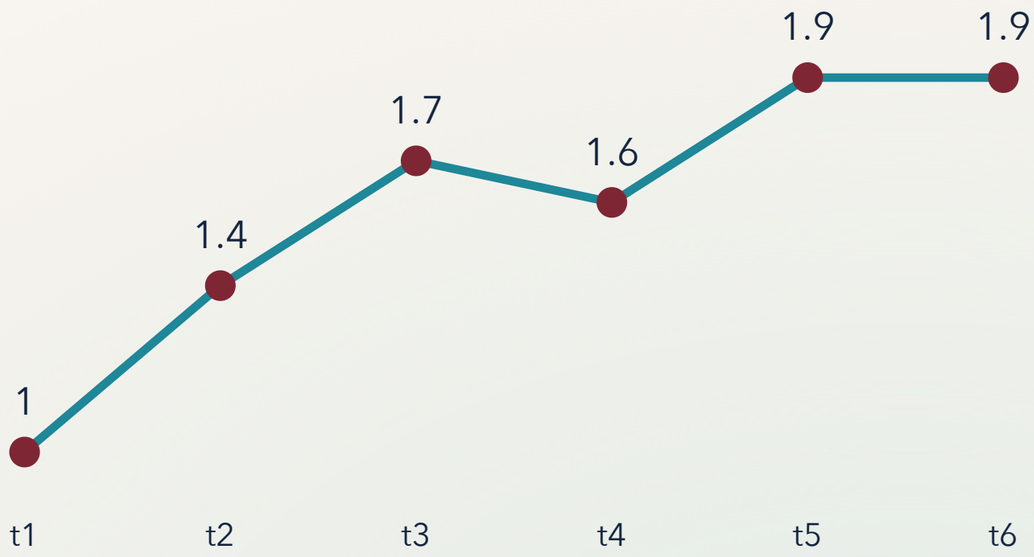


Mechanism diagram / illustrative teaching values

Persistence is a serious baseline

Predict the next observation using the last known observation.

$$\hat{y}[t] = x[t-1]$$



Order is part of the input

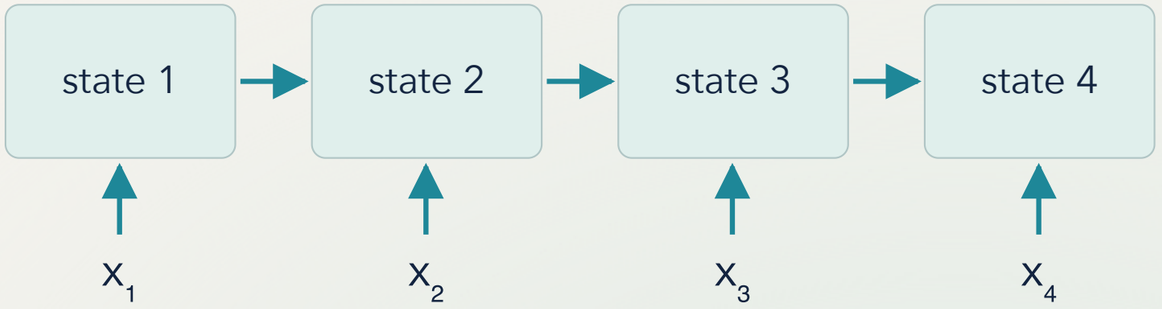
Mechanism diagram / illustrative teaching values

A recurrent state summarizes the past

An RNN applies the same state-update parameters at every time step.

$$h_t = \tanh(W_x x_t + W_h h_{t-1} + b)$$

Shared update parameters across time



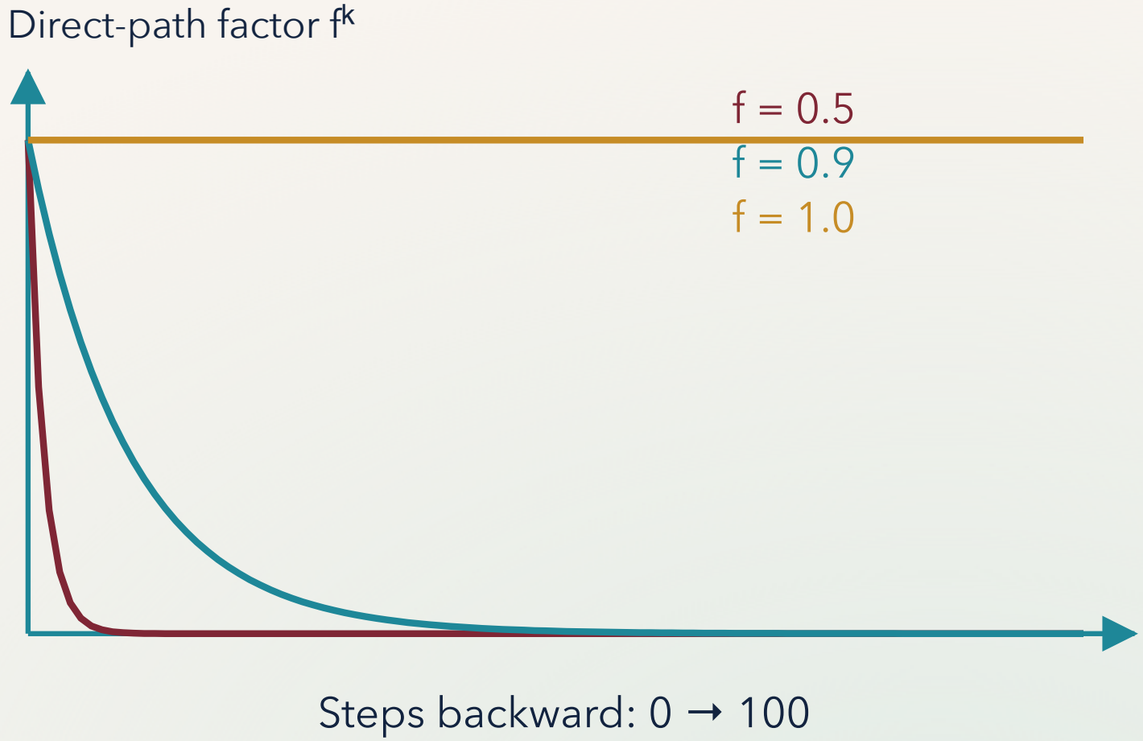
Past → running state → later state

Mechanism diagram / illustrative teaching values

Learning must cross the time chain

Backpropagation through time multiplies Jacobians along earlier states.

$$\partial h_t / \partial h_{t-k} = J_t \, J_{t-1} \, \dots \, J_{t-k+1}$$

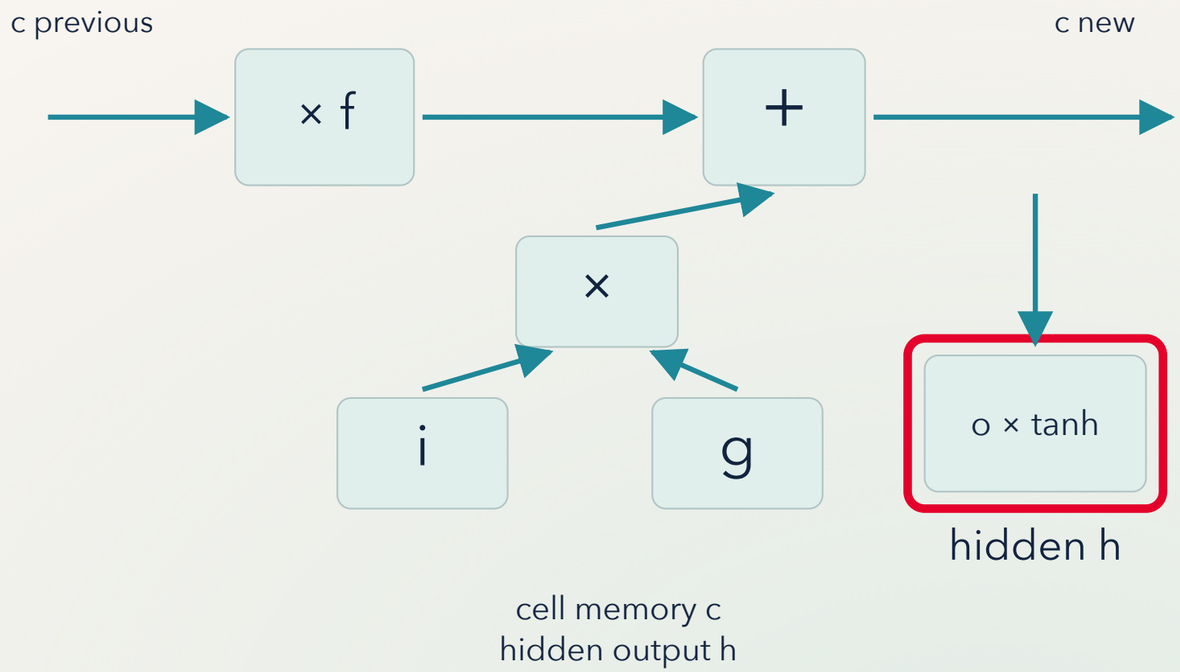


Mechanism diagram / illustrative teaching values

LSTM adds a separate memory path

The cell state carries memory; the hidden state exposes a filtered view.

> cell: c_t hidden output: h_t

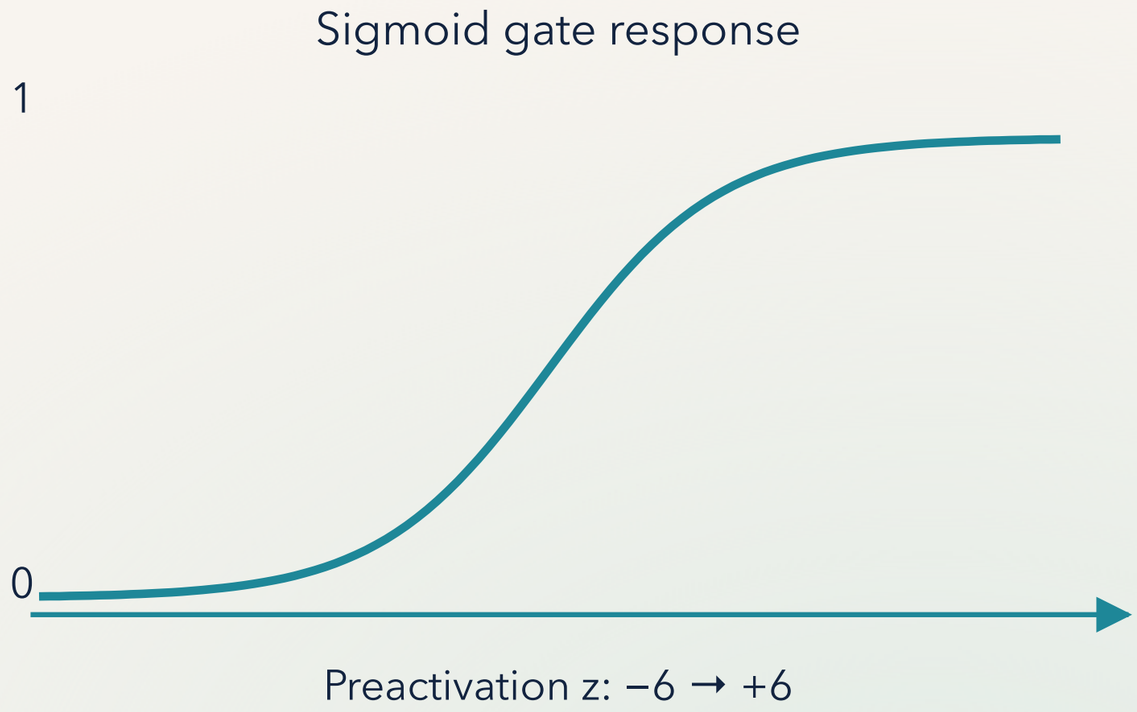


Mechanism diagram / illustrative teaching values

A gate is a learned vector of coefficients

Sigmoid outputs values between zero and one for each hidden dimension.

$$\sigma(z) = 1 / (1 + \exp(-z))$$

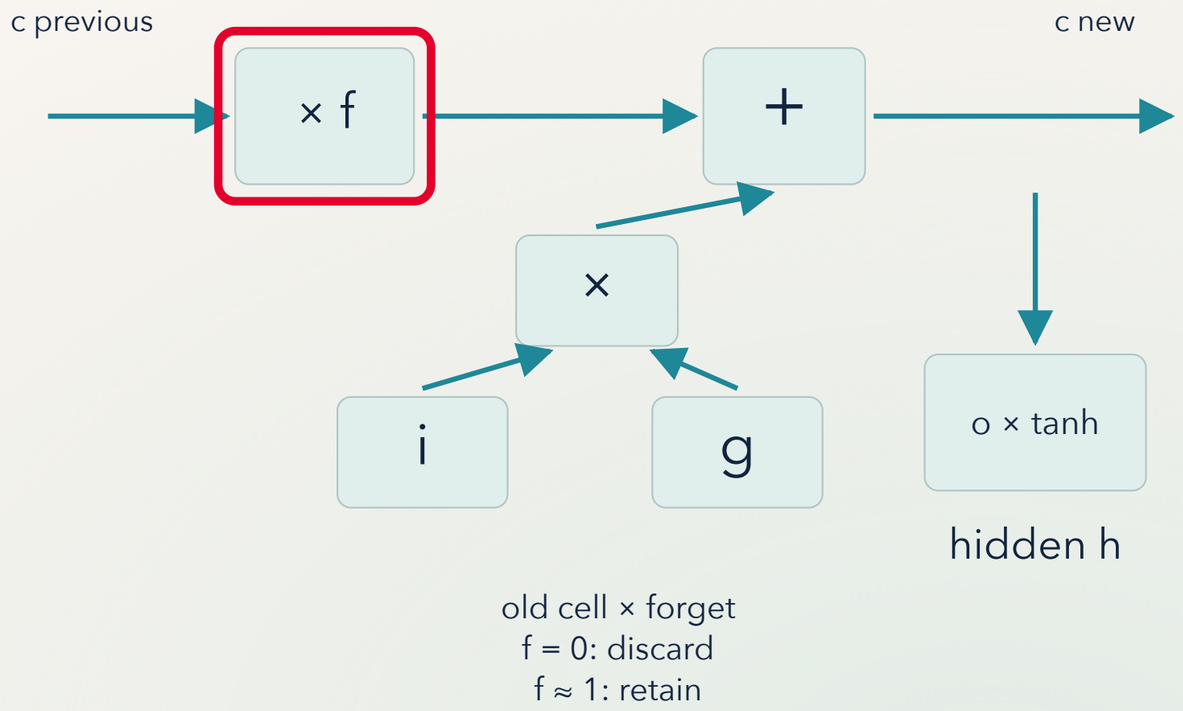


Mechanism diagram / illustrative teaching values

The forget gate controls retained memory

Multiply the previous cell state by a learned forget coefficient.

$$f_t = \sigma(W_f\,[h_{t-1}; x_t] + b_f)$$



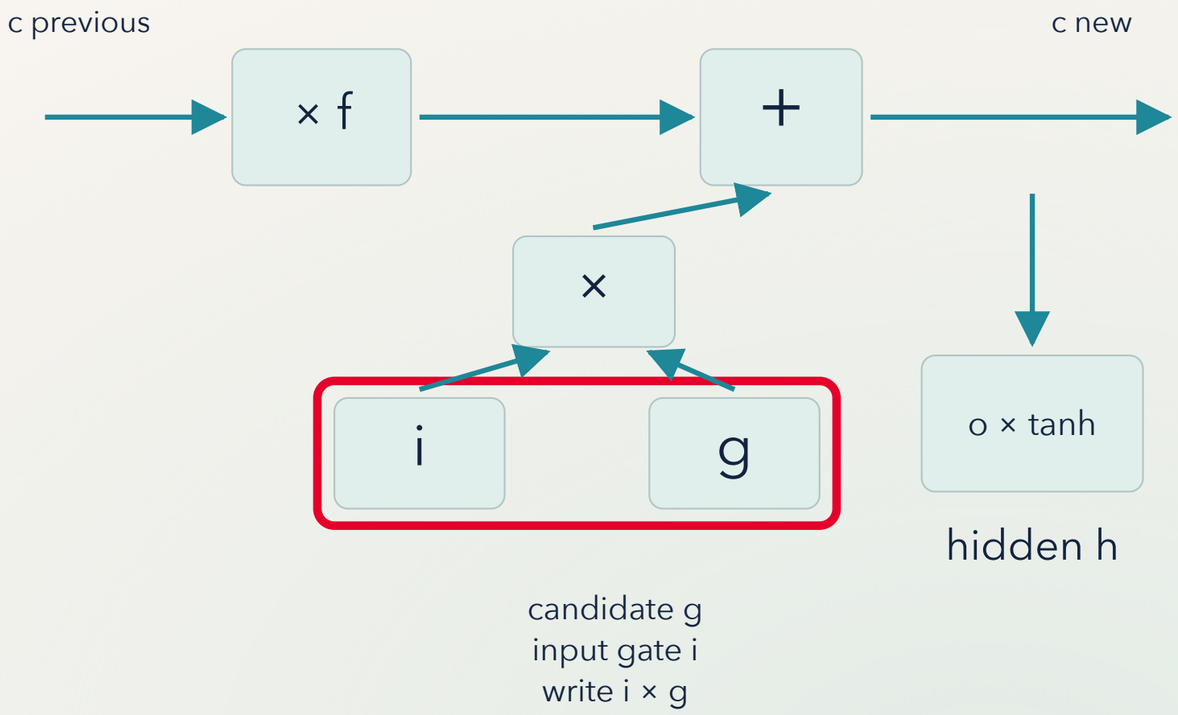
old cell × forget
f = 0: discard
f ≈ 1: retain

Mechanism diagram / illustrative teaching values

The input gate controls the write

A candidate proposes content; the input gate scales how much is written.

$$i_t = \sigma(W_i\,[h_{t-1}; x_t] + b_i) \quad g_t = \tanh(W_g\,[h_{t-1}; x_t] + b_g)$$

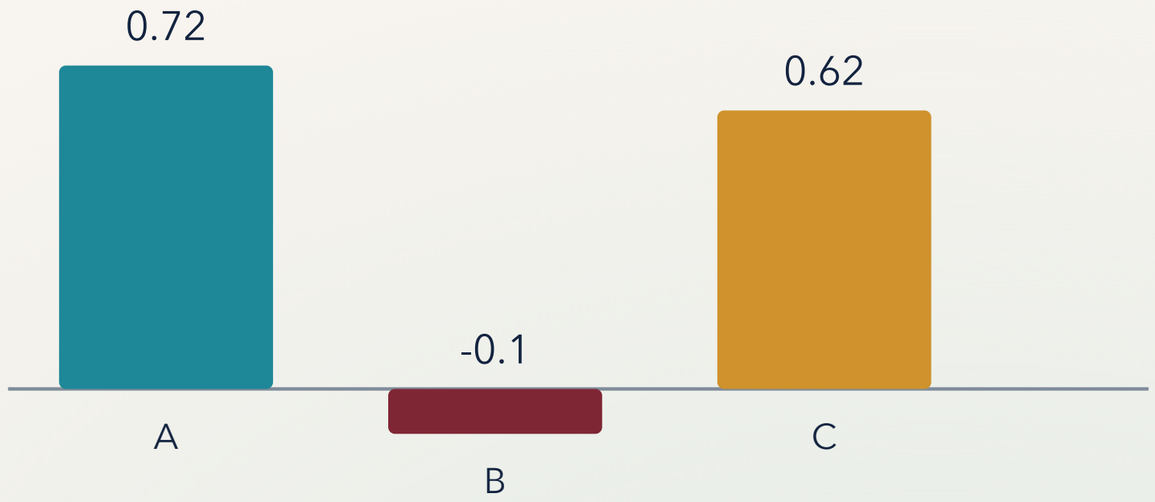


Mechanism diagram / illustrative teaching values

The cell update adds two paths

Retain selected old memory and add selected new content.

$$c_t = f_t \odot c_{t-1} + i_t \odot g_t$$



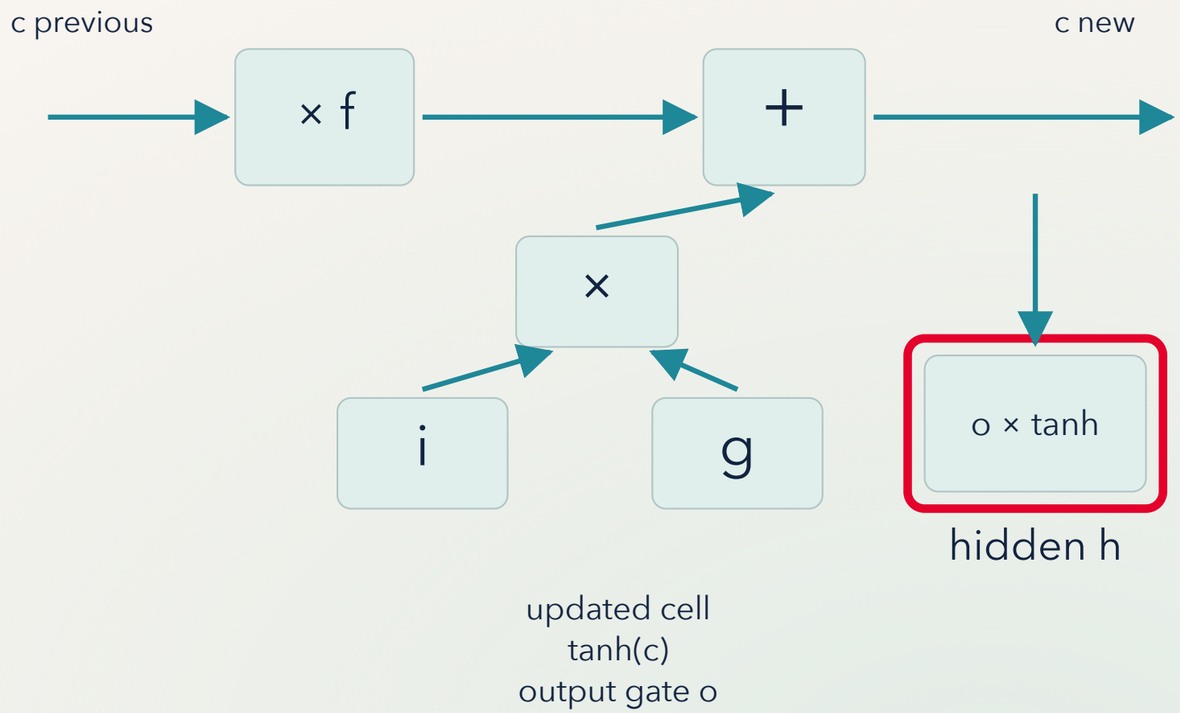
Illustrative values; see the stated calculation

Mechanism diagram / illustrative teaching values

The output gate exposes memory

The hidden state is a bounded, gated view of the updated cell.

> $o_t = \sigma(W_o\,[h_{t-1}; x_t] + b_o)$ $h_t = o_t \odot \tanh(c_t)$



Mechanism diagram / illustrative teaching values

Trace one numerical cell update

Chosen gates make every intermediate step visible.

$$0.9 \times 0.8 + 0.2 \times (-0.5) = 0.62$$

What is the hidden output?



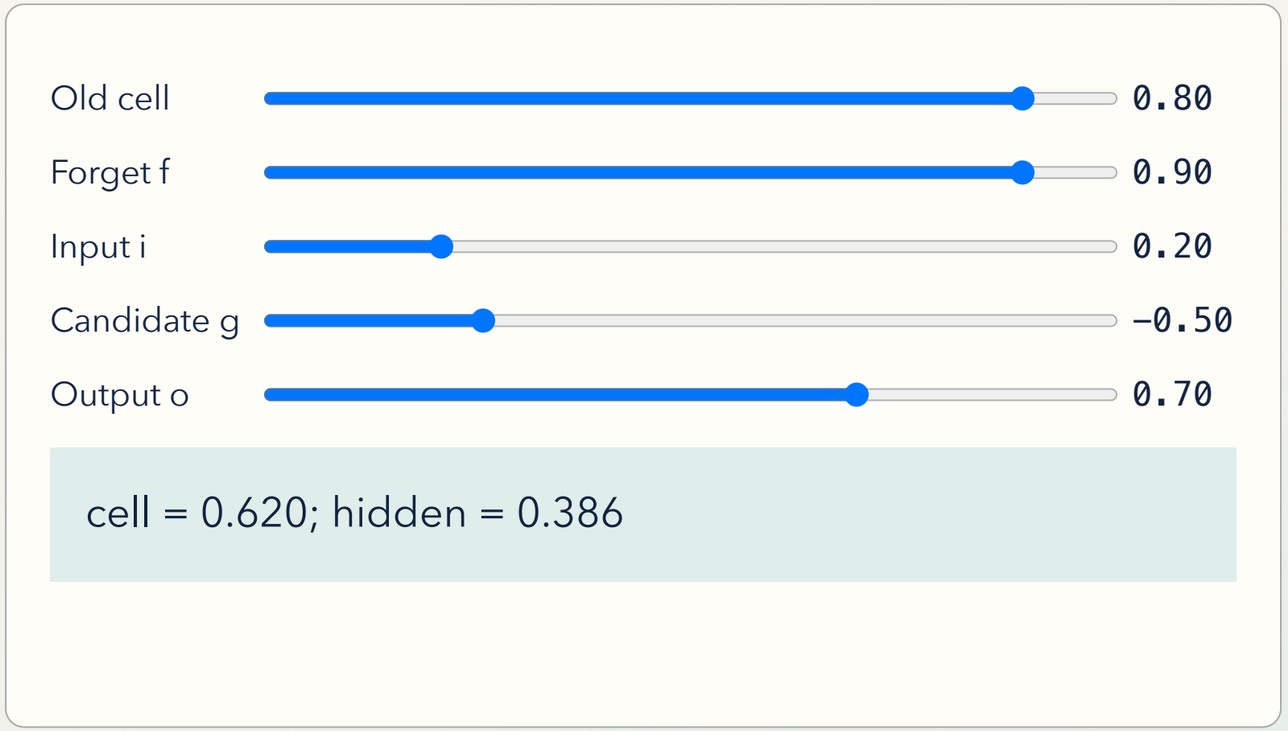
Illustrative values; see the stated calculation

Mechanism diagram / illustrative teaching values

Predict before moving a gate

A high forget coefficient does not prevent new content from changing memory.

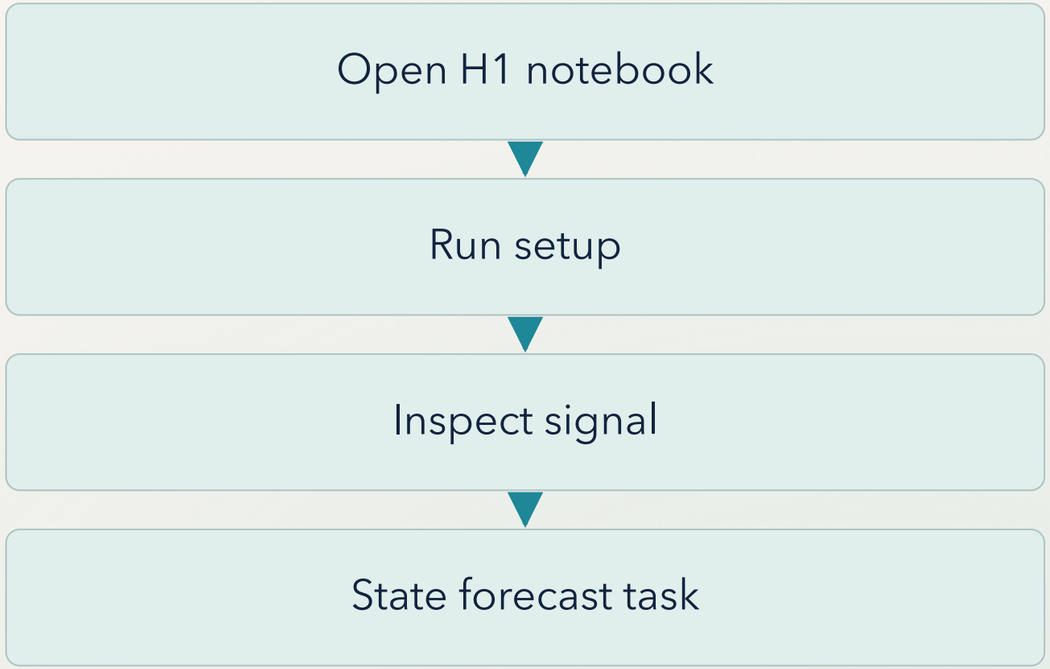
$$c_{new} = f \times c_{old} + i \times g$$



Interactive chosen values — change one variable

Open the sequence notebook

Run setup and inspect the synthetic sensor before fitting anything.

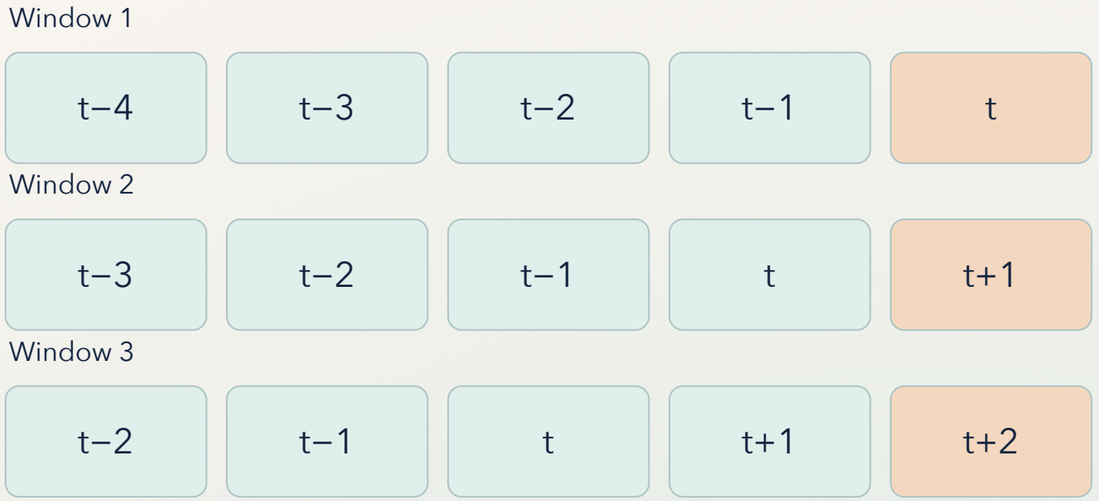


Mechanism diagram / illustrative teaching values

Check the window indices

A correct shape is insufficient if the window contains its target.

```
X[i] = signal[t−24:t] y[i] = signal[t]
```

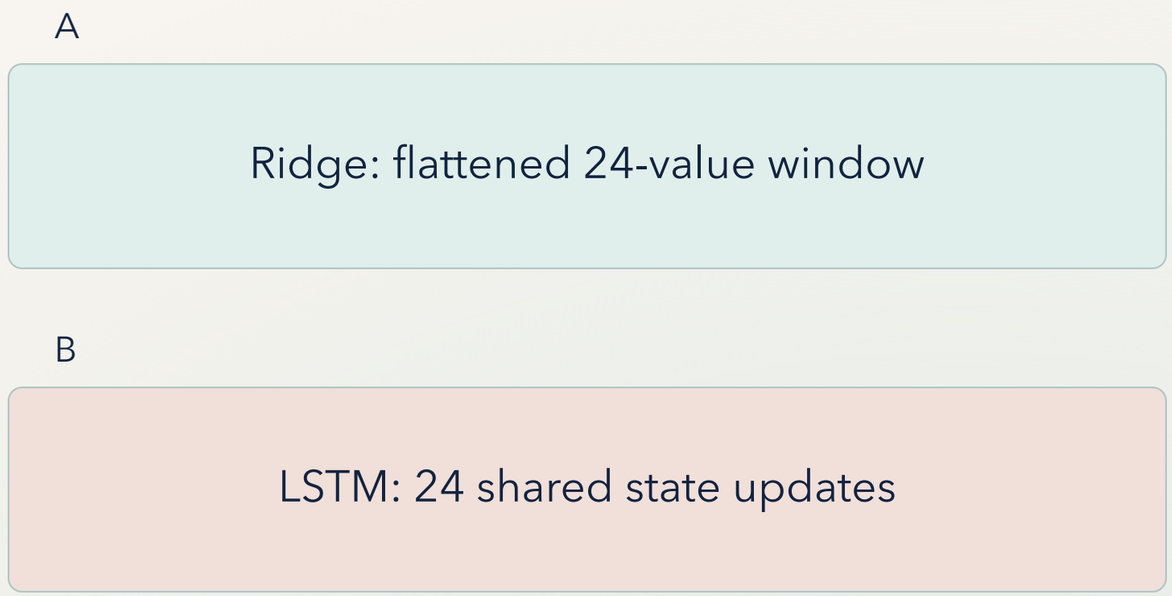


History shifts forward; target stays excluded

Mechanism diagram / illustrative teaching values

Compare linear memory with learned state

Ridge uses fixed window positions;
LSTM updates a shared recurrent state.

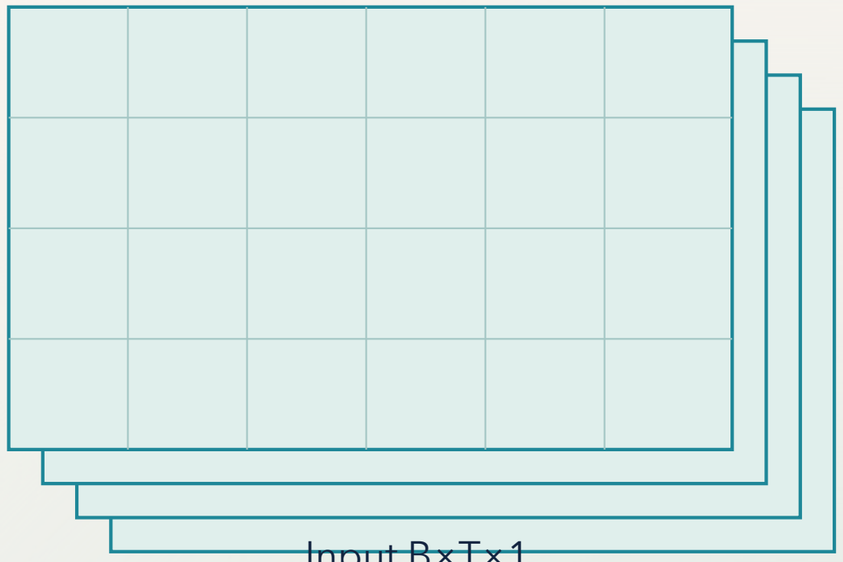


Mechanism diagram / illustrative teaching values

Map the implementation to tensor axes

batch_first changes input/output layout, not hidden-state layout.

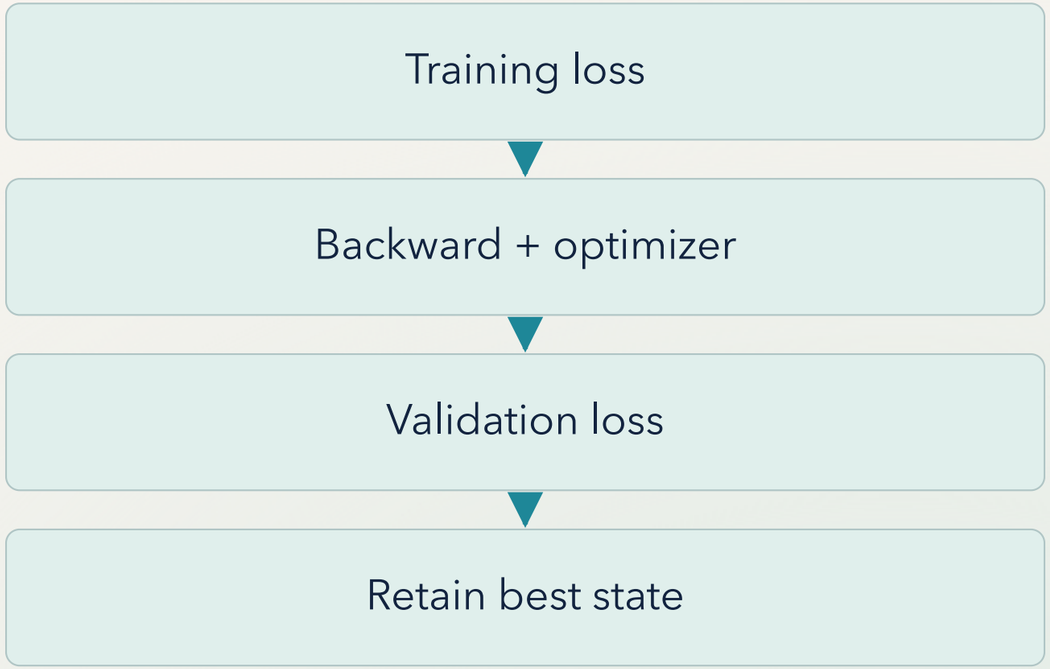
> `input: (B,T,1) states: (B,T,H) hₙ: (layers,B,H)`



Mechanism diagram / illustrative teaching values

Training and selection use different data

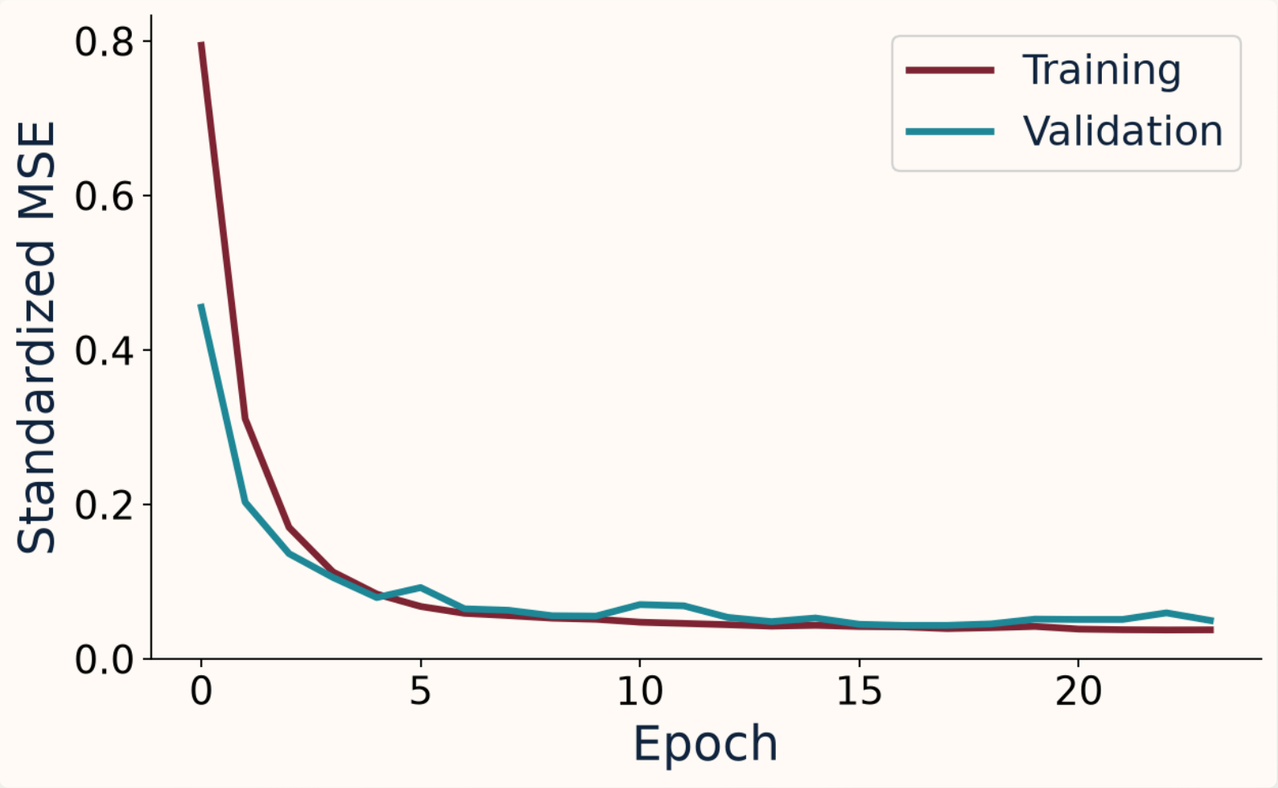
Gradient updates use training;
checkpoint selection uses validation.



Mechanism diagram / illustrative teaching values

A loss curve must come from the run

Inspect actual training and validation trajectories in your notebook.

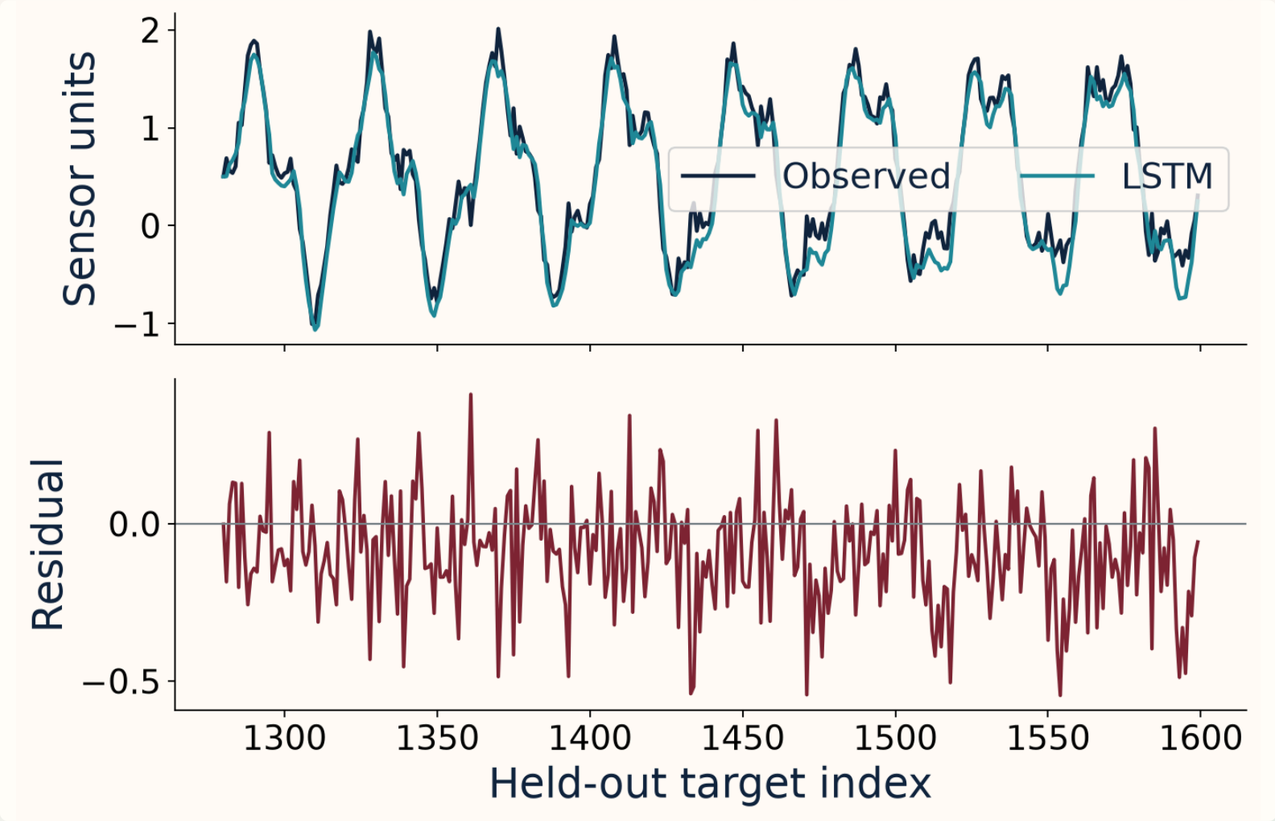


Actual CPU notebook run · synthetic teaching data

Judge errors in the original units

MAE averages error magnitude; RMSE emphasizes large residuals.

$$\mathrm{MAE} = \mathrm{mean}\ |\hat{y}-y| \qquad \mathrm{RMSE} = \sqrt{\mathrm{mean}(\hat{y}-y)^2}$$

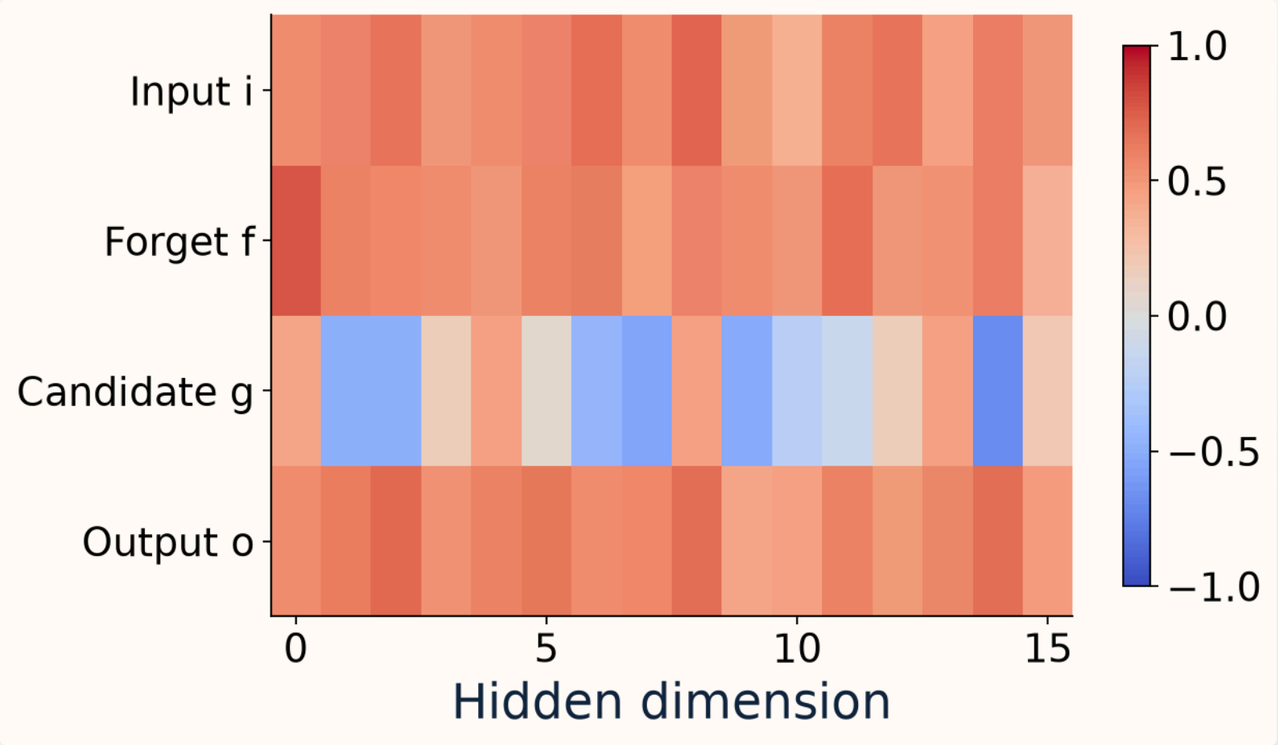


Actual CPU notebook run · synthetic teaching data

A trained gate can be reconstructed

PyTorch concatenates its learned gate parameters in i, f, g, o order.

> `linear projections → split → activations → cell update`



Actual CPU notebook run · synthetic teaching data

Reproduce one controlled change

Change window length while preserving the forecast question and split boundaries.

A

Reference: WINDOW = 24

B

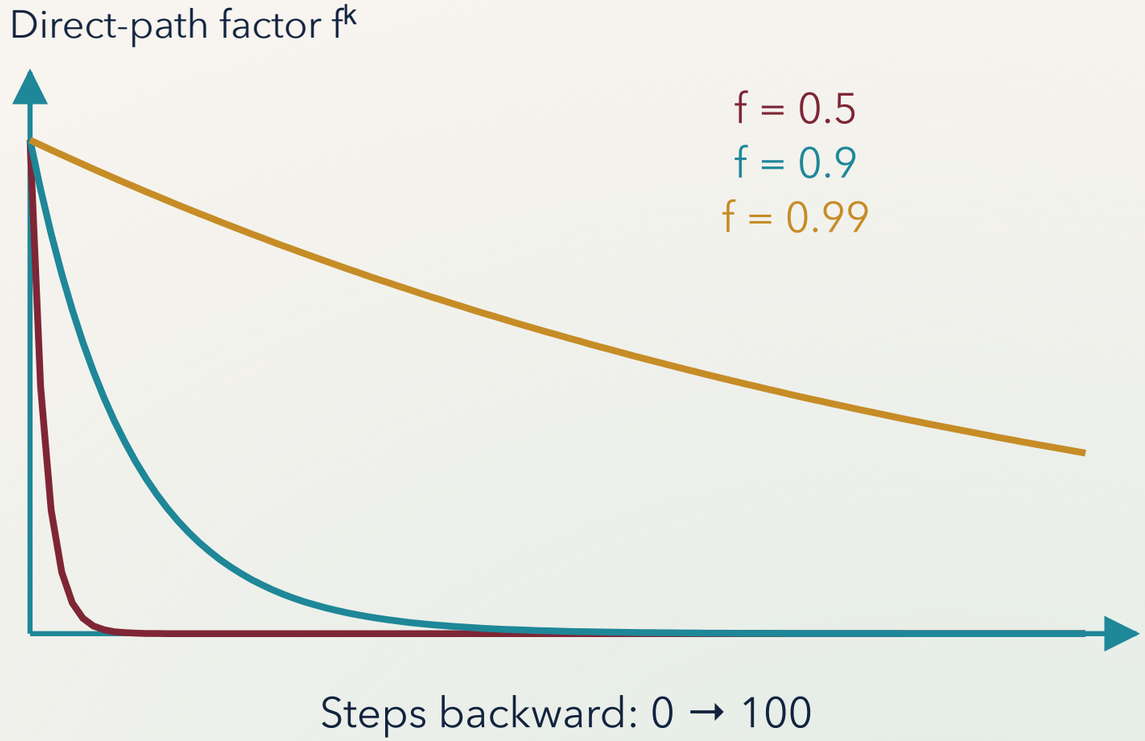
Intervention: WINDOW = 12 or 48

Mechanism diagram / illustrative teaching values

LSTM helps; it does not guarantee memory

Retention depends on learned gates, sequence structure, and optimization.

> direct memory factor ≈ $\prod f_j$

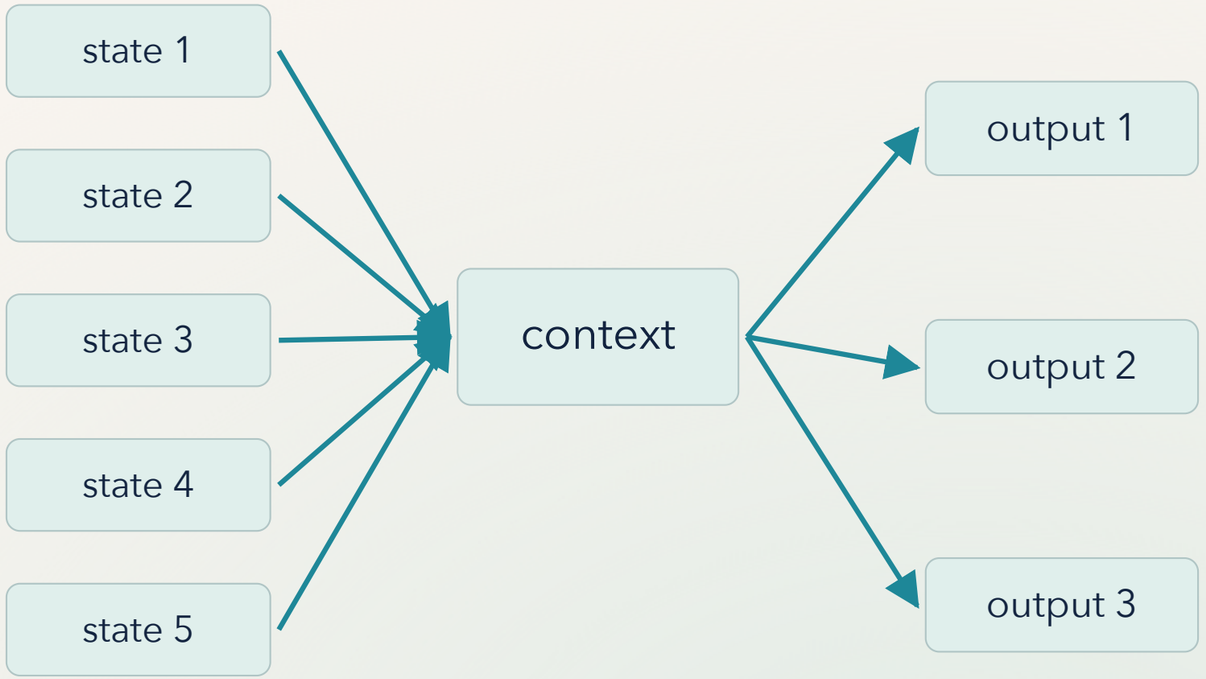


Mechanism diagram / illustrative teaching values

Exit: explain the compression problem

A final-state encoder compresses many inputs into one fixed-width vector.

What should H2 let each output retrieve?



Which information reaches each output?

Mechanism diagram / illustrative teaching values